Examiner only

Answer **all** *questions.*

1. *(a)* (i) State Ohm's law. [1]

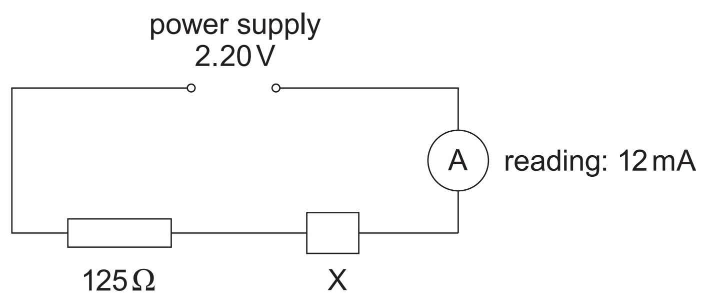
....

....

(ii) What can be said about the *resistance* of a conductor that obeys Ohm's law? [1]

....

(b) (i) An electrical component, X, is included in the circuit shown. The internal resistance of the power supply is negligible.



Show that the resistance of X in this circuit is approximately 60 Ω. [2]

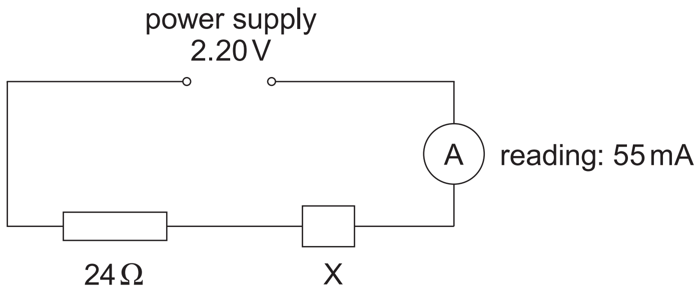
....

....

....

Examiner only

(ii) When the 125 Ω resistor in *(b)*(i) is replaced by a 24 Ω resistor, the reading on the ammeter increases, as shown below.



Evaluate whether or not X obeys Ohm's law, presenting your argument clearly. [2]

....

....

....

....

(iii) State, giving a reason, whether or not X could be a filament lamp. [1]

....

....

(c) A certain high temperature superconductor has a transition temperature of –188 °C. The boiling point of liquid nitrogen is –196 °C.

(i) State what is meant by the *transition temperature* of a superconductor. [1]

....

....

(ii) Give one possible use for a high temperature superconductor and state why it would be an advantage for the transition temperature to be above the boiling point of liquid nitrogen. [2]

....

....

....

....

10

2420U201 03

